MIL-V-5461C
12 May 1977
Superseding
MIL-V-5461B
6 April 1965

MILITARY SPECIFICATION

VALVE, VACUUM BREAKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.--This specification covers a vacuum breaking valve for engine-driven air pumps.

1.2 Classification.--Valve shall be of the following types, as specified (see 6.3):

Type I - 4.5 cubic feet per minute free air capacity.
Type II - 12 cubic feet per minute free air capacity.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-P-416 Plating, Cadmium (Electrodeposited)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Aviation Systems Command, P.O. Box 209, Main Office, St. Louis, Missouri 63166 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4820

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PPP-B-566	Boxes, Folding, Paperboard
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Fiberboard
PPP-B-676	Boxes, Set-Up, Paperboard

MILITARY

MIL-P-116	Preservation, Methods of
MIL-M-3171	Magnesium Alloy, Processes for Pretreatment and Preventive Corrosion Protection on
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-P-17667	Paper, Wrapping, Chemically Neutral (Non-Corrosive)

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MS33658	Fitting End, Hose Connection, Standard Dimensions for
MS87005	Valve, Vacuum Breaking

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publication.—The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Valve assembly.—Valve assembly shall be a vacuum-operated unit, provided with a means for external mechanical adjustment within the range specified in 3.3.4. Weight of Type I valve assembly shall not exceed 11 ounces and of Type II, 16 ounces (see 1.2). Dimensions shall be as specified on MS87005. Design and construction shall be simple. Valve assembly shall require only minimum maintenance.

3.1.1 Preproduction samples.—This specification requires preproduction testing. As soon as practicable after award of contract or order, and prior to submission of any complete units for final acceptance, two preproduction samples shall be subjected to the tests as specified in 4.2.1. A preproduction sample is defined as a production line sample of the product which the manufacturer will subsequently produce in fulfillment of the contract. The preproduction samples of the valve assembly shall be manufactured to the greatest extent possible by the use of tools, fixtures, and processes which will be used in the manufacture of additional valve assemblies on contract or order. Approval of the preproduction samples by the activity concerned shall not relieve the contractor of his responsibility to supply equipment conforming to this specification. Any changes or deviations from the preproduction samples shall be subject to the approval of the contracting officer.

3.2 Materials.—Materials shall be as specified herein. Materials which are not specifically specified shall be of standard commercial quality for this type of equipment, and shall be selected in accordance with MIL-STD-143.

3.2.1 Metals.—Metals shall be of the corrosion-resistant type, or protected to resist corrosion as follows:

a. Steel parts shall be cadmium-plated in accordance with Type I, Class 2 of QQ-P-416.

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b. Aluminum and aluminum alloy parts shall be anodized in accordance with MIL-A-8625.

c. Magnesium parts shall be treated in accordance with Type IV of MIL-M-3171.

Unless protected against electrolytic corrosion, dissimilar metals, as defined in MIL-STD-889, shall not be used in intimate contact with each other.

3.2.2 Castings.—Castings shall be clean, sound, and free from blow holes, porosity, cracks, and other defects.

3.3 Construction.—Suction relief air shall be routed through a #32 mesh, or finer, integral screen of rustproof construction. The valve construction shall withstand without damage, the normal strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.3.1 Hose connections.—Hose connections shall conform to MS33658.

3.3.2 Threads.—Straight screw threads shall be in accordance with MIL-S-7742. Pipe threads shall be in accordance with MIL-P-7105.

3.3.3 Locking.—All internal or external threaded parts shall be positively locked or safetied.

3.3.4 Adjustment.—The valve shall provide for adjustment (see 3.1) by discernible increments not more than 0.2-inch of mercury, within the calibration limits of plus or minus 3 inches of mercury from the nominal capacity. A stop shall be provided to prevent the adjustment from locking the valve shut. Provisions shall be incorporated for securing the adjustment in the position selected. The mechanical adjustment shall not require the use of special tools.

3.3.5 Lubrication.—The valve shall not require lubrication during its normal service life.

3.4 Performance.—The valve shall equal, or surpass, the minimum requirements established by the following tests:

a. Pressure drop (4.4.2).
b. Calibration (4.4.3).
c. Pressure fluctuations (4.4.4).
d. Leakage (4.4.5).
e. Endurance (4.4.6).
f. Extreme temperatures (4.4.7).

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3.5 Identification of product.—Valve shall be identified in accordance with MIL-STD-130.

3.6 Workmanship.—Workmanship shall insure uniformity of each valve produced under the contract or order. Finish shall be smooth, free of burrs, sharp edges, nicks, gouges, and scratches. Particular attention shall be given to cleanliness during fabrication and assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.—Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of examinations and tests.—Examinations and tests shall be classified as (1) preproduction and (2) conformance.

4.2.1 Preproduction.—The two preproduction samples (see 3.1.1.) shall be subjected to all tests specified in 4.4. Test report in accordance with MIL-STD-831 shall be submitted to, and approved by, the contracting officer prior to production.

4.2.2 Conformance.—Conformance examinations and tests (see 4.3) shall consist of examinations to determine compliance with this specification with respect to design, construction, marking, and workmanship, and all tests specified in 4.4.2 through 4.4.5.

4.3 Sampling.—From each lot submitted, sample valves shall be selected at random in accordance with MIL-STD-105. Inspection and conformance tests shall be at level II.

4.3.1 Lot.—A lot shall consist of all valves of the same part number which have been manufactured under essentially the same conditions.

4.3.2 Acceptable quality level.—Acceptable quality level (AQL) for major and minor defects shall be as specified in Table I.

4.4 Tests.—

4.4.1 Methods.—Valves shall be tested with a test setup as shown on Figure 1, or equivalent, using the pump designated for the type valve

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specified thereon. Unless otherwise specified, pressures and suctions are gage values. Temperatures shall be 60° to 80°F for all tests except those specified in 4.4.7 and 4.4.8. Test valve integral screen shall not be removed.

4.4.2 Pressure drop.--The pressure drop (H) across the valve shall not exceed 1.5 inches of mercury when measured with the valve set for maximum suction and tested with the pump operated at speeds required to draw 1/4, 1/2, 3/4, and maximum test airflow, as specified for types I and II while the inlet pressure (P_1) is maintained at 26 inches of mercury absolute. Maximum test airflow shall be as follows:

Type I valve	12 cubic feet per minute free air.
Type II valve	30 cubic feet per minute free air.

4.4.3 Calibration.--Procedures for calibration tests shall be as follows:

4.4.3.1 Setting No. 1.--Set the discharge pressure (P_2) at 1 inch of mercury at 1,500 revolutions per minute (rpm). Set the valve suction (P_3) at 6.1 inches, plus or minus .1-inch, of mercury, when the flow through the airflow meter is 25 percent of the nominal rated capacity as specified below, and the air pump is driven at 1,500 rpm. Maintain the rate of flow through the airflow meter and run the pump at 2,250, 3,000, 3,750, and 4,125 rpm. The suction (P_3) shall not deviate from the original setting by more than 0.5-inch of mercury. Nominal rated capacity of valves shall be as follows:

Type I valve	4.5 cubic feet per minute free air.
Type II valve	12 cubic feet per minute free air.

4.4.3.2 Setting No. 2.--Set the discharge pressure (P_2) at 1 inch of mercury at 1,500 rpm. Draw all the air through the screen and set the suction relief valve suction (P_3) at 4.5 inches, plus or minus 0.1-inch, of mercury, at a speed of 1,500 rpm. Run the pump at 2,250, 3,000, 3,750, and 4,125 rpm. The suction (P_3) shall not deviate from the original setting by more than 0.75-inch of mercury.

4.4.3.3 Setting No. 3.--Repeat procedure of setting No. 2 with suction (P_3) at 7.0 inches, plus or minus 0.2-inch, of mercury. The suction (P_3) shall not deviate from the original setting by more than 0.5-inch of mercury.

4.4.4 Pressure fluctuations.--The suction (P_3) shall not fluctuate more than 0.1-inch of mercury during the calibration tests (see 4.4.3).

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TABLE I. — Classification of defects in accordance with MIL-STD-105.

Categories	AQL percent defective	Defects	Paragraph reference
CRITICAL:	None defined		
MAJOR:	4.0		
101		Nonconformance with MS87005	3.1
102		Valve exceeds maximum weight	3.1
103		Materials not corrosion resistant or not protected to resist corrosion	3.2.1
104		Dissimilar metals not protected and in intimate contact with each other	3.2.1
105		Casting cracked	3.2.2
106		Screen size larger than specified	3.3
107		Hose connections not as specified	3.3.1
108		Incorrect threads (pipe bosses)	3.3.2
109		Valves improperly identified	3.5
110		Failure to pass any of the following tests:	
		(a) Pressure drop	4.4.2
		(b) Calibration	4.4.3
		(c) Pressure fluctuation	4.4.4
		(d) Leakage	4.4.5
MINOR:	6.5		
201		Materials not selected in accordance with MIL-STD-143, but otherwise suitable	3.2
202		Threads burred or contaminated with metal particles	3.3.2
203		Adjustment increments cannot readily be determined by turning of the external mechanical adjustment	3.3.4
204		Threaded parts not locked or safetied	3.3.3
205		Valve exterior scratched, gouged, porous, nicked, has burrs or sharp edges	3.6
206		Valve interior contaminated with metal particles or otherwise dirty	3.6

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4.4.5 Leakage.—The leakage (difference in cubic feet per minute measured in run 1 and run 2 under the following conditions) shall not exceed 5 percent of the nominal rated capacity of the valve specified in 4.4.3.1:

(a) Pump speed 1,500 rpm.
(b) Discharge pressure (P_2) 1 inch of mercury.
(c) Suction relief valve inlet suction (P_3) 4 inches of mercury as controlled by the hand valve in the inlet line.
(d) Suction relief valve settings:
Run 1 - Plug the screen opening airtight.
Run 2 - Set for a suction (P_3) of 5 inches of mercury, plus or minus 0.1-inch, when all the air is drawn through the screen and the pump is running at 1,500 rpm.

4.4.6 Endurance.—The endurance test setup shall conform to Figure 1, except that the airflow meter may be replaced by a special screened fitting which will pass the nominal rated capacity of the valve at a suction of 6 inches of mercury. The outlet pressure (P_2) may be any value and the inlet suction (P_3) shall be 6 inches of mercury, plus or minus 0.2-inch. The pump shall be run as specified below, drawing the nominal rated airflow through the special screened fitting or airflow meter and all excess air through the filter or screen. No part of the valve shall fail.

Length of run (hours)	Pump speed rpm, plus or minus 100 rpm
50	1,500
50	2,250
200	3,750
50	4,125

4.4.6.1 Recalibration.—Following the endurance test, the valve shall be recalibrated and shall satisfy the leakage and calibration test requirements (see 4.4.3 and 4.4.5).

4.4.7 Extreme temperatures.—

4.4.7.1 Low temperature.—The valve shall be conditioned at a temperature of minus 54° C., plus or minus 1° C. (minus 65° F., plus or minus 2° F.), for at least 2 hours. The leakage test shall be repeated while at this temperature. Provision shall be made to pre-cool the air to the specified temperature during the leakage test. The temperature of the valve shall then be allowed to change rapidly to room temperature while the leakage test is repeated.

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4.4.7.2 High temperature.--Procedure specified in 4.4.7.1 shall be repeated at 71° C., plus or minus 1° C. (160° F., plus or minus 2° F).

4.4.8 Salt spray.--A salt spray test shall be conducted as specified in MIL-E-5272. Valve ports shall not be sealed.

4.5 Preparation for delivery.--Packaging, packing, and marking shall be inspected to determine conformance with Section 5 of this specification.

5. PACKAGING

5.1 Preservation and packaging.--Preservation and packaging shall be level A or C, as specified (see 6.3).

5.1.1 Level A.--Each valve shall be packaged in accordance with Method III of specification MIL-P-116. The valve shall be wrapped in chemically neutral paper conforming to Type I of MIL-P-17667 and placed in a snug-fitting container conforming to weather-resistant class of PPP-B-636 or conform to variety 2 of PPP-B-566 or PPP-B-676.

5.1.2 Level C.--Valves shall be packaged in a manner to afford protection against damage during shipment from the supply source to the receiving activity.

5.2 Packing.--Packing shall be level A, B, or C, as specified (see 6.3). Insofar as possible, shipping containers shall contain an identical number of items, shall be of a uniform size, and shall snugly enclose the contents.

5.2.1 Level A.--Valves packaged as specified in 5.1 shall be packed in an overseas type, style optional, grade A container conforming to PPP-B-601.

5.2.2 Level B.--Valves packaged as specified in 5.1 shall be packed in fiberboard boxes conforming to weather-resistant class of PPP-B-636.

5.2.3 Level C.--Valves packaged as specified in 5.1 shall be packed in a manner to insure carrier acceptance and safe delivery to destination.

5.3 Marking.--In addition to any special marking required by the contract or order, unit packages and shipping container shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use.--The vacuum breaking valve is intended for use

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with engine-driven air pump assemblies, to prevent damage to the air pump due to stoppage of the air inlet line, and to regulate the suction in the air inlet line.

6.2 National stock numbers.—National stock numbers of valves are as follows:

Type I as shown on MS87005-1	NSN 4820-00-513-6405
Type II as shown on MS87005-2	NSN 4820-00-593-5096

6.3 Ordering data.—Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type valve required (see 1.2).

(c) Level of preservation, packaging, and packing required (see 5.1 and 5.2).

(d) Whether special marking is required (see 5.3).

Custodians:

Army - AV
Navy - AS
Air Force - 99

Review activities:

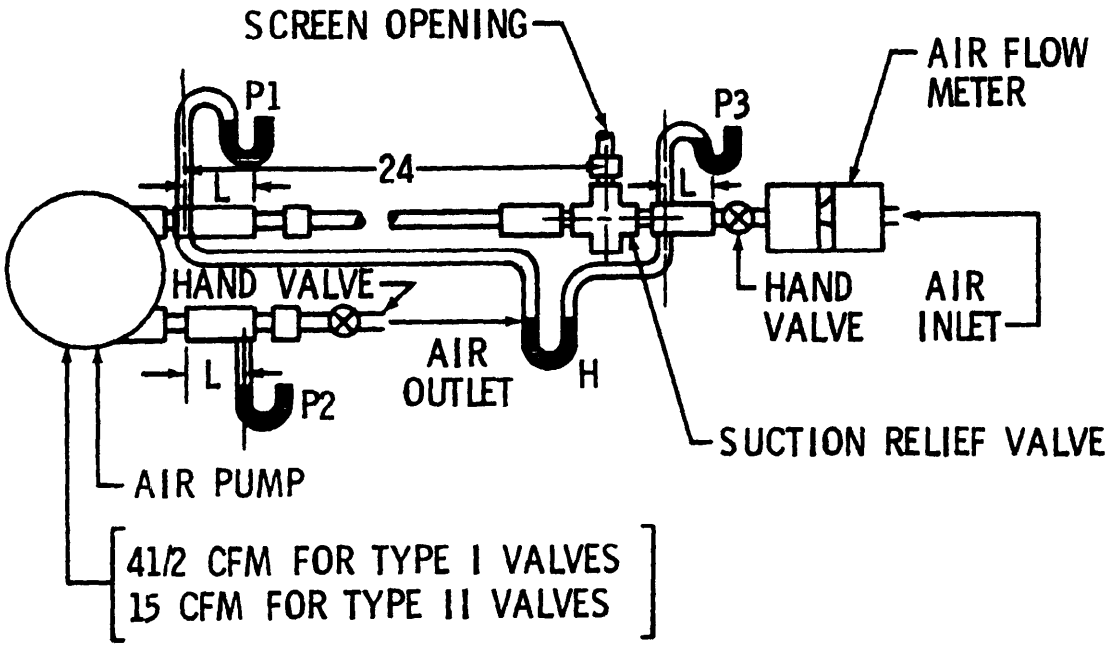
Navy - AS
Air Force - 99
DSA - CS

Preparing activity:

Army - AV

Project No. 4820-0309

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POINT OF MANOMETER CONNECTION SHALL BE PROCEDED BY A STRAIGHT SECTION OF FITTING OF MINIMUM LENGTH "L" AND "ID" AS SHOWN BELOW:

DIAMETER OF PUMP PORT OPENING	INSIDE DIAMETER OF FITTING	MINIMUM LENGTH OF FITTING "L"
3/8	.42	2-1/2
1/2	.54	3
5/8	.65	4
3/4	.90	5-1/2
1	1.15	7

FIGURE 1. AIR PUMP VACUUM-BREAKING VALVE TEST SETUP.

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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